

Pedestrian Crossing Intention Prediction From Surveillance Videos for Over-the-Horizon Safety Warning

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Abstract—Pedestrian crossing intention prediction could effectively prevent traffic injuries and improve pedestrian safety. This paper focuses on pedestrian crossing intention prediction from surveillance cameras, which could provide over-the-horizon safety warnings and has the potential to better ensure pedestrian safety, compared with that from on-board cameras. However, most prediction-based methods are designed with a fundamental assumption that the visual data is collected from an on-board camera rather than a bird-eye-view one, thus the prevalent methods in this research domain do not match surveillance scenarios. To deal with this issue, an automated learning framework is proposed, in which a pedestrian-centric environment graph is primarily constructed to reflect visual variations and spatiotemporal relationships between pedestrians and their surroundings. After that, a Graph Convolutional Network (GCN) based environment encoder and a pedestrian-state encoder are designed to extract prominent environment features and pedestrian behavior features, respectively. Finally, an intention prediction decoder is developed to extrapolate the probability of crossing intention. Experimental results demonstrate that each component in the framework contributes to performance improvement and their combination obtains state-of-the-art performance, suggesting the effectiveness and superiority of our framework.

Index Terms—Traffic safety, pedestrian crossing intention prediction, surveillance scenario, GCN, environment graph.

I. INTRODUCTION

PEDESTRIAN crossing intention prediction, which refers to predicting whether a pedestrian attempts to cross the street in the future seconds, is of great significance for preventing traffic injuries and improving pedestrian safety [1]. With the advancement of computer vision techniques, understanding and anticipating the behaviors of pedestrians standing on the curbside have attracted many researchers' attention, especially those focusing on autonomous vehicles. Accurately and timely predicting pedestrian crossing intention could provide ample

time for drivers to respond to underlying crashes, effectively preventing collision and enhancing safety for vulnerable road users (e.g., pedestrians).

To date, an increasing number of studies have engaged in the task of pedestrian crossing intention prediction based on on-board cameras [2], [3], which are mainly designed for Advanced Driver Assistance Systems (ADAS). Thereinto, early methods mainly focus on exploiting pedestrian states (e.g., history trajectories [4], [5] and/or skeleton data [6], [7]) to recognize their crossing intention. Nevertheless, considering only the pedestrian states is not sufficient, as other auxiliary cues such as ego-vehicle dynamics (e.g., speed, acceleration) and environmental information (e.g., zebra, stop sign) are important for crossing intention prediction as well. For instance, a vehicle at high speed would change the pedestrian's intention to cross. In consideration of the importance of such auxiliary cues, some recent studies [2], [8] explore crossing intention via modeling pedestrian states, vehicle dynamics and environmental information into an encoder-decoder network, significantly improving the detection performance.

It is undeniable that those methods can ensure the safety of pedestrians crossing the street by releasing accurate early warnings. However, the view fields of on-board cameras are generally limited and the deep networks adopted in those studies are oftentimes exceedingly sophisticated and over-parameterized, which may slow down the computational speed and result in the untimely release of early warnings. In contrast, surveillance cameras are widely distributed on urban roads and could provide high-volume and high-view monitoring information [9], [10]. In the context of Connected Vehicle (CV), perception information from surveillance cameras could be timely transmitted to any vehicles in the road network after being processed by cloud computing devices with stronger computing power, as shown in Figure 1 (a). This offers the potential to convey the over-the-horizon warnings of pedestrian crossing intention to vehicles of interest in a timely manner, thereby granting drivers sufficient reaction time. As such, this paper will focus on pedestrian crossing intention prediction from surveillance cameras instead of on-board cameras.

Despite the relative success of existing on-board camera-based methods that incorporate both pedestrian states and auxiliary cues, those methods cannot be readily applied to surveillance scenarios due to disparities in observable environmental information. For example, on-board camera-based

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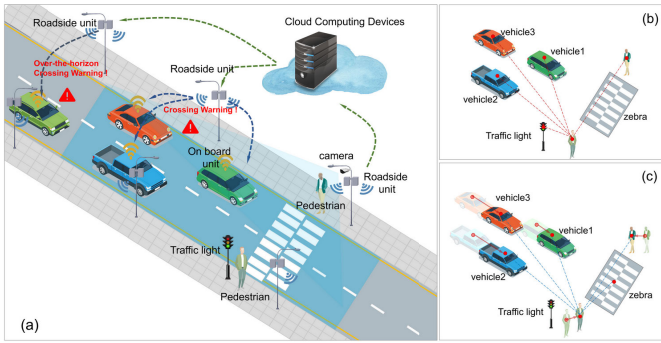


Fig. 1. (a) Schematic diagram of crossing intention prediction from surveillance cameras; (b) an appearance sub-graph is constructed to reflect appearance variations (e.g., the color change of traffic light); (c) a motion sub-graph is constructed to represent spatiotemporal relationships between pedestrians and their surroundings (e.g., approaching vehicles).

methods typically consider only the spatial correlation between pedestrians and ego-vehicles, while in surveillance scenarios, the spatial correlation may need to be extended to that with several nearby vehicles, pedestrians and/or traffic infrastructure due to the expansion of perception range. As such, it motivates us to propose a more suitable crossing intention prediction framework for surveillance scenarios.

To this end, an automated learning framework is proposed, which learns to comprehend the intentions or underlying motives of pedestrians through incorporating multiple information from both pedestrians and their surroundings. Primarily, a pedestrian-centric environment graph consisting of an appearance sub-graph and a motion sub-graph is constructed to reflect visual variations (e.g., the color change of traffic light) and spatiotemporal relationships between pedestrians and their surroundings (e.g., approaching vehicles), as shown in Figure 1 (b-c). Based on the environment graph, a GCN-based environment encoder is designed to extract prominent environment features that strongly influence the pedestrian crossing intention. Moreover, a pedestrian-state encoder is adopted to embed three types of pedestrian states (i.e., history trajectory, pedestrian appearance, and skeleton data) into unified features that highly abstract pedestrian's motion and appearance information. After feature extraction, an intention prediction decoder is developed to extrapolate the probability of crossing intention.

To sum up, our contributions in this study are as follows:

(1) Targeting the challenging pedestrian crossing intention prediction from surveillance videos, we design a novel automated learning framework to leverage intrinsic cues and mutual relations underlying both pedestrians and their surroundings to figure out their crossing intentions.

(2) A pedestrian-centric environment graph is constructed to represent appearance variations and spatiotemporal relationships between pedestrians and their surroundings, based on which a GCN-based environment encoder is designed to extract spatiotemporally correlated features that are conducive to crossing intention prediction from the perspective of pedestrians' environment.

(3) A pedestrian-state encoder is proposed to encode three types of pedestrian states (i.e., history trajectory,

pedestrian appearance, and skeleton data) into features that highly abstract pedestrian's motion and appearance information, which are conducive to crossing intention prediction from the perspective of pedestrians themselves.

II. RELATED WORKS

Pedestrian crossing intention prediction falls under the umbrella of pedestrian action prediction, which aims to predict whether a pedestrian will cross the road in the next short time [1]. The pedestrian crossing intention could be extrapolated from both the states of pedestrians themselves (e.g., history trajectories and body poses) and their environments (e.g., traffic signals, vehicle-pedestrian and curbside-pedestrian distances). As such, existing methods could be divided into three categories according to features utilized to predict crossing intention, i.e., trajectory prediction-based, 2D/3D pose-based and scene modeling-based methods.

A. Trajectory Prediction-Based Methods

Early works predict pedestrian trajectories through some parametric methods, based on which pedestrians' crossing intention could be derived with acceptable accuracy. Those methods including Kalman Filter (KF) [11], Gaussian Process Dynamical Models (GPDMs) and B-GPDM [12] generally utilize position, motion and/or action information of pedestrians to predict the future trajectory.

Another approach for pedestrian trajectory prediction and intention prediction is based on deep learning models. Thereinto, Rasouli et al. [2] formulate the pedestrian intention prediction as a binary classification problem, whereby an RNN encoder-decoder is constructed to estimate the pedestrian crossing intention and predict future trajectories. Kotseruba et al. [1] devise a 3D convolutional branch and multiple parallel individual RNN branches to effectively incorporate implicit (e.g., local context information) and explicit features (e.g., pose, location, and ego-vehicle speed). Other methods, such as Social LSTM [4] and Social GAN [5] and Social-STGCNN [13], take into consideration the interaction between pedestrians and thus attain promising performance for trajectory prediction.

B. 2D/3D Pose-Based Methods

As outlined in [7], the pedestrian's pose can be used as a strong indicator of its intention to cross the road. Thus, some works have been proposed to predict crossing intention based on pedestrians' 2D/3D poses. Fang and López [6] first hypothesize that pedestrians' walking patterns and postures determine whether they have the intention to cross. Then, a detector-tracker pipeline along with a pose estimation module is used to obtain the pedestrians' skeletons, followed by a Random Forest (RF) classifier applied to predict the crossing intention of vulnerable road users (VRUs). Zhang et al. [14] divide pedestrians' states into three categories and apply four machine learning models to predict pedestrians' red light crossing intention. Gesnouin et al. [15] introduce a multi-branch predictor named TrouSPI-Net to predict the crossing behaviors

of pedestrians based on pseudo-image sequences of skeletal joints' positions. Instead of using 2D skeleton/pose data, Xu et al. [16] propose a 3D pose estimation model based on graph convolutional network (GCN) to recognize the actions of traffic police, cyclists and pedestrians in traffic scenes.

C. Scene Modeling-Based Methods

Modeling both scene information and pedestrian states could boost the prediction performance but is more challenging compared with those only modeling trajectories or poses. Several recent studies have tried to bridge pedestrians' states and their environment to infer the intensity of crossing intention. Hariyono and Jo [17] propose a walking human model to identify and localize pedestrians who are crossing the road according to their poses, locations, relative distances from vehicles and other local context information. Yang et al. [18] propose a pedestrian crossing intention recognition (PCIR) framework, in which the local traffic scene is made full consideration by a distance encoding module. Concerning the issue that different features might not be well fused, Yang et al. [19] introduce a hybrid fusion strategy to fuse the spatio-temporal features of global and local context information.

More recently, GCNs have been applied in pedestrian trajectory prediction and intention prediction to model the complicated interactions between pedestrians and ego-vehicle as well as environment. For example, Chen et al. [3] use a GCN-based autoencoder to encode the relationship between pedestrians and the approaching ego-vehicle.

The aforementioned methods have considered many factors that may affect pedestrian crossing intention and achieved impressive performance. However, those methods are designed for autonomous driving systems, which tries to anticipate the near future from the limited view fields of on-board cameras. In contrast, surveillance cameras could provide over-the-horizon view fields for drivers, which are more suitable for pedestrian crossing intention prediction and safety warning. Nonetheless, existing methods cannot be directly transferred to surveillance scenarios that have broader view and more environmental factors influencing the prediction of pedestrian crossing intention. For example, existing methods only bridge the spatial correlations between pedestrians and ego-vehicles, while in surveillance scenarios, such spatial correlations may need to be extended to several nearby vehicles, pedestrians and/or traffic infrastructure. As such, this study will fill this gap and devise a framework customized for surveillance scenarios.

III. PROPOSED FRAMEWORK

A. Motivation

As outlined in [2], pedestrian crossing intention prediction refers to anticipating whether a pedestrian i has the intention to cross the street $int_i \in \{0, 1\}$ after a short period of observation (e.g., 0.5s in [2]). Therefore, we define the task as a binary classification problem in this study.

Prior to designing our framework, we first discuss and analyze how the administrators of surveillance systems determine the intensity of pedestrian crossing intention. It is intuitive

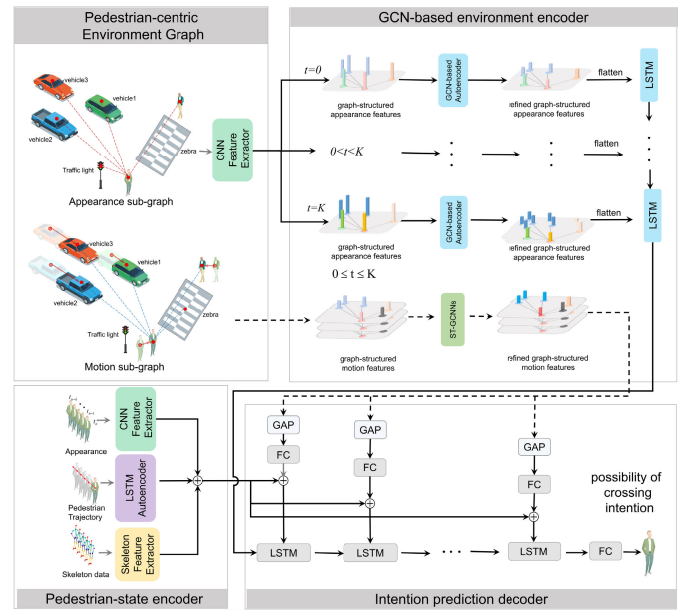


Fig. 2. Overall network architecture of the proposed pedestrian crossing intention prediction framework. Specifically, the input of the framework are three types of pedestrian states (i.e., history trajectories, pedestrian appearances, and skeleton data) and a pedestrian-centric environment graph consisting of an appearance sub-graph and a motion sub-graph. The two sub-graphs provide the information about visual variations (e.g., a traffic light is turning green) and spatiotemporal relations (e.g., a vehicle is approaching to the pedestrian), respectively. After feature extraction and aggregation, an intention prediction decoder is proposed to extrapolate the possibility of crossing intention.

that the administrators might examine pedestrians' historical trajectory, behavior/posture, and appearance (such as whether they are observing their surroundings or engaged with their cell phones) to extrapolate the crossing intention possibilities from the perspective of pedestrians themselves. Then, they may survey the surrounding environment to assess whether the environmental condition is allowed for pedestrian crossing. Finally, the possibilities of pedestrian crossing intention will be estimated by simultaneously taking the above two steps into consideration. Drawing inspiration from this process, we develop a novel automated learning framework that utilizes intrinsic cues and mutual relationships between pedestrians and their surroundings to accurately predict their crossing intentions.

B. Overview

The overall architecture of the proposed framework is demonstrated in Figure 2. The framework takes as input three types of pedestrian states (i.e., history trajectory, pedestrian appearance, and skeleton data) and a pedestrian-centric environment graph that consists of an appearance sub-graph and a motion sub-graph. The two sub-graphs provide the information about visual variations (e.g., a traffic light is turning green) and spatiotemporal relations (e.g., a vehicle is approaching to the pedestrian), respectively. To extract more representative features underlying those inputs, a GCN-based environment encoder and a pedestrian-state encoder are proposed.

In the GCN-based environment encoder, graph-structured appearance and motion features inferred from the appearance and motion sub-graphs are taken as inputs. Specifically,

a GCN-based autoencoder is built to refine the appearance features, followed by feature association in the temporal dimension through a long short-term memory (LSTM) network. Meanwhile, a Spatio-Temporal Graph Convolutional Neural Network (ST-GCNN) is exploited to build spatiotemporal relations from graph-structured motion features and uncover underlying social relations between pedestrians and their surroundings.

In the pedestrian-state encoder, three types of pedestrian states (i.e., history trajectories, pedestrian appearances, and skeleton data) are taken as inputs. A CNN feature extractor, an LSTM autoencoder, and an ST-GCNN feature extractor are used to extract features related to pedestrians' appearances, trajectories and skeletons, respectively. After that, the three types of features are aggregated into an unified pedestrian-state feature. All resulting features from the two encoders are fed to an LSTM-based intention decoder, which further extrapolates the possibility of crossing intention.

The proposed framework differs from existing methods in two key ways. Firstly, regarding the modeling of mutual relations, existing methods (e.g., Social LSTM and Social STGCNN) typically employ pooling operations or graph neural networks to model the motion interaction between homogeneous traffic participants (i.e., pedestrians). In contrast, this paper models both the motion and appearance interactions between heterogeneous traffic elements (i.e., traffic participants and infrastructure) by constructing an ST-GCNN and a GCN autoencoder, respectively. Secondly, regarding the modeling of intrinsic cues, existing methods often rely on pedestrian poses or trajectories to infer potential crossing intentions embodied by pedestrians themselves. However, this paper takes a more comprehensive approach by simultaneously considering the poses, trajectories, and appearances of pedestrians, in order to fully explore the intrinsic cues that influence pedestrian crossing intention. In the following subsections, we will provide detailed explanations of the inputs and networks used in the proposed framework.

C. Pedestrian-Centric Environment Graph

To represent visual variations and spatiotemporal relationships between pedestrians and their surroundings, a pedestrian-centric environment graph consisting of an appearance sub-graph and a motion sub-graph is constructed. Thereinto, the appearance sub-graph reflects the visual variations of pedestrian-related traffic elements (e.g., vehicles, traffic lights, and other pedestrians), while the motion sub-graph builds the spatiotemporal relations between pedestrians and their surroundings through embedding the spatiotemporal information (i.e., location, velocity and acceleration) into graph nodes.

1) *Appearance Sub-Graph*: Pedestrians' surroundings are likely to affect their crossing intention. For example, a pedestrian may attempt to cross the road when the corresponding sidewalk signal is turning green or the slowdown warning light of an approaching vehicle is lightened. Those visual variations may affect pedestrian crossing intentions in part, so in this study they are taken into account to improve the prediction performance of our framework.

Formally, given an observed pedestrian and a set of M vehicles, N pedestrians and L traffic signals in the vicinity at each time step t , we construct a pedestrian-centric appearance sub-graph G_A^t representing the visual features of traffic elements extracted by a convolutional neural network. Concretely, G_A^t is defined as $G_A^t = (V_A^t, E_A^t)$, where $V_A^t = \{v_i^t | \forall i \in \{1, \dots, M+N+L+1\}\}$ is the set of vertices of G_A^t . For each vertex in G_A^t , its attribute is assigned to the visual feature of corresponding instance that is extracted by a pre-trained classification network (e.g., GhostNet [20]). To note, we adopt the first fully connected layer of the classification network to attain the visual feature of size (1×1280) for each traffic element in our study. E_A^t is the set of edges of G_A^t , where $E_A^t = \{e_{ij}^t | \forall i, j \in \{1, \dots, M+N+L+1\}\}$. e_{ij}^t equals 1 if v_i^t and v_j^t are connected, and equals 0 otherwise. Considering the difference of connection strength between each pair of nodes, a kernel function a^t is used to endow different importance to each edge e_{ij}^t and generate a weighted adjacency matrix of appearance sub-graph, which is denoted as A_d^t . The kernel function a^t is defined as:

$$a_{ij}^t = \begin{cases} \left\| (x_i^t, y_i^t) - (x_j^t, y_j^t) \right\|_2^{-1}, & \text{if } \left\| (x_i^t, y_i^t) - (x_j^t, y_j^t) \right\|_2 \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where (x_i^t, y_i^t) and (x_j^t, y_j^t) are the center coordinates of the bounding boxes for instances i and j . The closer (x_i^t, y_i^t) and (x_j^t, y_j^t) are, the larger a_{ij}^t will be and the greater importance to e_{ij}^t will be endowed.

2) *Motion Sub-Graph*: Apart from visual variations, spatiotemporal relationships between pedestrians and their surroundings also influence their crossing intentions. For example, pedestrians approaching the crosswalk are more likely to cross the street than those just passing by. Also, a pedestrian may interrupt its intended desire to cross when a vehicle is rapidly approaching, or rise up its crossing intention when surrounding pedestrians attempt to cross. Those phenomena suggest that the spatiotemporal relationships are also of great significance for pedestrian crossing intention prediction. To this, we take in account the factors reflecting spatiotemporal relationships between pedestrians and their surroundings, such as the distance variation to nearby zebra crossing, or the relative speed and acceleration of nearby vehicles.

Slightly different from the assumption in the appearance sub-graph, we assume that there are M vehicles, N pedestrians and Z zebra crossings in the vicinity of the observed pedestrian at each time step t . Based on this, we devise a pedestrian-centric motion sub-graph $G_M^t = (V_M^t, E_M^t)$. Here, $V_M^t = \{v_i^t | \forall i \in \{1, \dots, M+N+Z+1\}\}$ is a set of vertices and $E_M^t = \{e_{ij}^t | \forall i, j \in \{1, \dots, M+N+Z+1\}\}$ is a set of edges. For each vertex in V_M^t , its attribute is assigned to category information (e.g., pedestrian, vehicle or zebra crossing), motion information (e.g., direction, velocity and acceleration) and position information (e.g., corner coordinates of bounding boxes and the shortest distance to zebra crossing) calculated from two or several consecutive frames.

Formally, suppose that the history bounding boxes of a pedestrian i are denoted as $\{(x_{i,l}^t, y_{i,l}^t, x_{i,r}^t, y_{i,r}^t) | t = 1, 2, \dots, K\}$ and the position of the nearest zebra crossing j is denoted as $(x_{j,l}, y_{j,l}, x_{j,r}, y_{j,r})$, where the subscripts l and r represent the upper left and lower right corners, and x and y represent the abscissa and ordinate in the pixel coordinate system. Specifically, the motion information is defined as:

$$x_{i,lm}^t = \frac{x_{i,l}^t + x_{i,r}^t}{2} \quad (2)$$

$$y_{i,lm}^t = y_{i,r}^t \quad (3)$$

$$v_x^t = x_{i,lm}^t - x_{i,lm}^{t-1} \quad (4)$$

$$v_y^t = y_{i,lm}^t - y_{i,lm}^{t-1} \quad (5)$$

$$a_x^t = v_x^t - v_x^{t-1} \quad (6)$$

$$a_y^t = v_y^t - v_y^{t-1} \quad (7)$$

$$\alpha = \arctan \frac{y_{i,lm}^t - y_{i,lm}^{t-1}}{x_{i,lm}^t - x_{i,lm}^{t-1}}, \quad (8)$$

where $x_{i,lm}^t$ and $y_{i,lm}^t$ denote the abscissa and ordinate of the bottom center of the bounding box $(x_{i,l}^t, y_{i,l}^t, x_{i,r}^t, y_{i,r}^t)$; v_x^t, v_y^t denote the velocities along the x-axis and y-axis at time t ; a_x^t and a_y^t denote the accelerations; α represents the motion direction.

The pedestrian's distance to zebra crossing, which could be regarded as the shortest distance from an outer point to a rectangle, is defined as the minimum distance from the pedestrian to all sides and corners of the bounding box of the zebra crossing:

$$d_{i,j}^t = \begin{cases} \min \left(\left\| y_{i,lm}^t - y_{j,s}^t \right\|_1 \right), s \in \{r, l\} \text{ if } x_{j,l}^t \leq x_{i,lm}^t \leq x_{j,r}^t \\ \min \left(\left\| x_{i,lm}^t - x_{j,s}^t \right\|_1 \right), s \in \{r, l\} \text{ if } y_{j,l}^t \leq y_{i,lm}^t \leq y_{j,r}^t \\ \min \left(\left\| (x_{i,lm}^t, y_{i,lm}^t) - (x_{j,s_1}^t, y_{j,s_2}^t) \right\|_2 \right), \\ s_1, s_2 \in \{r, l\} \text{ others} \end{cases} \quad (9)$$

After calculating the motion information (i.e., $v_x^t, v_y^t, a_x^t, a_y^t$ and α) and the position information (i.e., $x_{i,l}^t, y_{i,l}^t, x_{i,r}^t, y_{i,r}^t$ and $d_{i,j}^t$), the attribute of each vertex in V_M^t consists of a 13-dimensional feature vector, in which category information, motion information and position information are stored in three, five and five elements, respectively.

For each edge e_{ij}^t in E_M^t , it equals 1 if v_i^t and v_j^t in V_M^t are connected, and equals 0 otherwise. Moreover, a weighted adjacent matrix of motion sub-graph (denoted as A_M^t) is generated, considering that two vertices with closer spatial distance are more likely to influence each other. The specific definition of kernel function at time step t is consistent with that defined in formula (1).

D. GCN-Based Environment Encoder

After constructing the two environment sub-graphs, their respective encoders are designed for feature extraction. Graph neural networks are effective methods for modeling graph-structured data in the non-Euclidean domain [21].

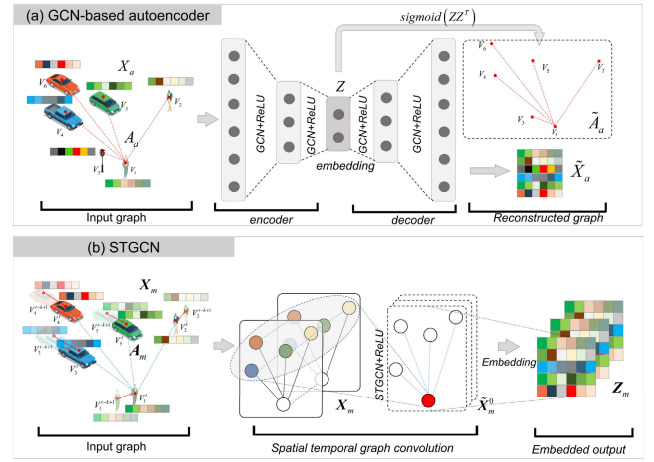


Fig. 3. Model architectures of (a) the GCN-based autoencoder and (b) the ST-GCNN. In the GCN-based autoencoder, two layers of GCN and ReLU are used as encoder and another two layers are as decoder. The encoder and the decoder are trained in an unsupervised manner. In the ST-GCNN, a single layer of STGCN, which operates one graph convolution and three temporal convolutions, is adopted to extract motion features reflecting spatiotemporal relationships between pedestrians and their surrounding.

Therefore, we construct the environment encoders based on graph neural networks to extract generalized features representing pedestrian crossing intention-related visual variations and spatiotemporal relationships. Respectively, a GCN-based autoencoder is built to refine features inferred from the appearance sub-graph, followed by feature association in the temporal dimension through a long short-term memory (LSTM) network. Also, a Spatio-Temporal Graph Convolutional Neural Network (ST-GCNN) is exploited to build spatiotemporal relations from graph-structured motion features and uncover underlying dynamics relations between pedestrians and their surroundings.

1) *GCN-Based Autoencoder*: As mentioned in Section III-C.1, a classification network is introduced to obtain the visual feature of size (1×1280) for each traffic element, so as to construct the appearance sub-graph. In our assumption, there are a total of K frames given to predict pedestrian crossing intention, and each of them includes an observed pedestrian and a set of M vehicles, N pedestrians and L traffic signals nearby. In this case, the input size of the LSTM network will be $K \times (M + N + L + 1) \times 1280$, which may not only consume a lot of computing resources but also hinder the LSTM from learning spatiotemporally correlated representations effectively. To decrease the computational cost and facilitate LSTM learning, a GCN-based autoencoder is proposed to refine the graph-structured appearance features and its model architecture is demonstrated in Figure 3 (a).

To improve upon existing GCN-based autoencoders such as VGAE [22] and GALA [23], which only learn to reconstruct either the graph structure or the node attributes, we develop a new GCN-based autoencoder that simultaneously considers both reconstruction objectives. This approach allows the autoencoder to more effectively learn a spatiotemporally correlated representation for the appearance sub-graph. Formally, the graph structure and the node attribute matrix of the appearance sub-graph are assumed as A_a and X_a , respectively.

According to [24], the layer propagation of graph convolution is defined as:

$$X_a^{(l+1)} = \text{ReLU} \left(\hat{D}^{-\frac{1}{2}} \hat{A}_a \hat{D}^{-\frac{1}{2}} X_a^{(l)} W^{(l)} \right), \quad (7)$$

where $\hat{A}_a = A_a + I$ and $\hat{D}$ is the diagonal node degree matrix of $\hat{A}_a$. $X_a^{(l)}$ represents the node attribute matrix inputted to l -th GCN with internal parameters $W^{(l)}$, and ReLU is a common activation function called Rectified Linear Unit.

In this paper, two layers of GCN and ReLU are used as the encoder and another two layers are as the decoder. Thereinto, the encoder embeds the original input X_a to a node embedding Z , with tensor shape varies from $(M+N+L+1, 1280)$ to $(M+N+L+1, 256)$. As the counterpart of the encoder, the decoder reconstructs a node attribute matrix $\tilde{X}_a$ that is approximate to X_a from the node embedding Z , with tensor shape varies from $(M+N+L+1, 256)$ back to $(M+N+L+1, 1280)$. Notably, the reconstructed graph structure is directly inferred from the node embedding Z , with a straightforward calculation $\tilde{A}_a = \text{sigmoid}(ZZ^T)$, where Z^T is the transpose matrix of Z . To enable the reconstructed graph structure $\tilde{A}_a$ and the node attribute matrix $\tilde{X}_a$ approximate to the original graph structure A_a and node attribute matrix X_a , a reconstruction loss is devised to train the autoencoder, namely,

$$L_{rec} = \gamma \cdot \|\tilde{A}_a - A_a\|_F^2 + (1 - \gamma) \cdot \|\tilde{X}_a - X_a\|_F^2, \quad (8)$$

where $\|\cdot\|_F^2$ denotes the Frobenius norm, and $0 \leq \gamma \leq 1$ is a hyper-parameter that balances the weight between the reconstruction losses of graph structure and node attribute matrix. After the autoencoder's optimization, the well-trained encoder is used to refine features inferred from the appearance sub-graph and meanwhile decrease the computational cost of the overall framework.

2) *Spatio-Temporal Graph Convolutional Neural Network*: As mentioned in section III-C.2, spatiotemporal relationships between pedestrians and their surroundings also influence their crossing intentions, we thereby construct a motion sub-graph in the form of spatial-temporal graph. To extract more representative features from such a motion sub-graph, a Spatial-Temporal Graph Convolution Neural Network (ST-GCNN) is elaborately constructed, whose architecture is demonstrated in Figure 3 (b).

Unlike GCN-LSTM models [25], [26], which are commonly used GCN-based spatial-temporal models, ST-GCNN uses GCN to extract features across both spatial and temporal domains, thereby avoiding the parameter inefficiency and expensive cost of recurrent architectures (e.g., LSTM) [27]. Formally, the graph structure and the node attribute matrix of the motion sub-graph are assumed as A_m and X_m . The layer propagation of graph convolution is similar with that described in III-C.1, but with slight differences:

$$X_m^{(l+1)} = \text{ReLU} \left(\hat{D}_m^{-\frac{1}{2}} \hat{A}_m \hat{D}_m^{-\frac{1}{2}} X_m^{(l)} W_m^{(l)} \right), \quad (9)$$

where $\hat{A}_m = \{\hat{A}_m^{t-K+1}, \dots, \hat{A}_m^t\}$ is a stack of adjacent matrices from time step $t - K + 1$ to time step t , and $\hat{A}_m^t = A_m^t + I$; $\hat{D}_m^{-\frac{1}{2}}$ is the diagonal node degree matrix of $\hat{A}_m$, $X_m^{(l)}$ is the

TABLE I
SEVERAL CNN MODELS' PERFORMANCE ON THE IMAGENET DATASET ALONG WITH THEIR NUMBER OF PARAMETERS AND FLOATING POINT OPERATIONS

Model	Weights (M)	FLOPs (M)	Top-1 Acc. (%)
VGG-16[30]	138.4	1550	74.4
ResNet-50 [31]	25.6	410	75.3
ShuffleNetV1 [32]	1.9	138	67.8
ShuffleNetV2 [34]	2.3	146	69.4
MobileNetV1 [36]	1.3	150	63.3
MobileNetV2 [34]	2.2	141	66.7
MobileNetV3 [35]	4.0	155	73.3
GhostNet [20]	5.2	141	73.9

node attribute matrix that is inputted to the l -th ST-GCNN layer with internal parameter $W_m^{(l)}$.

E. Pedestrian-State Encoder

The previously mentioned contents mainly focus on extrapolating the crossing intention from the perspective of spatiotemporal relationships between pedestrians and their surroundings. However, we believe that the history states of the observed pedestrian are significant for crossing intention prediction as well. For example, pedestrians approaching the road with trajectories that do not decelerate oftentimes have stronger crossing intentions than those with decelerated trajectories. Additionally, pedestrians looking around are more likely to cross than those looking down at their phones. Motivated by those observations, we devise a pedestrian-state encoder that takes as inputs three types of pedestrian states (i.e., pedestrian appearances, history trajectories, and skeleton data). Specifically, three corresponding feature extractors (i.e., appearance, trajectory, skeleton feature extractors) are introduced to abstract the appearance, trajectory and skeleton features of the observed pedestrian, respectively. We will detail the three feature extractors in the following sub-sections.

1) *Appearance Feature Extractor*: Compared to traditional image processing-based algorithms such as SIFT [28] and SURF [29], Convolutional Neural Networks (i.e., CNN) could provide more robust appearance features for instances in images. As such, we adopt a CNN pretrained on the ImageNet dataset to extract features that highly abstract the pedestrians' appearance information (e.g., gaze, posture, and on-calling actions). Notably, we have introduced a CNN in Section III-C.1. To perform the feature extraction more efficiently, we adopt the identical CNN in this section. Moreover, in an observed video frame, the appearance features of all instances (e.g., vehicles, pedestrians, and traffic lights) are simultaneously captured according to their bounding boxes, after extracting the features of the whole frame at once.

Currently, there are many CNN models available for image feature extraction, such as conventional VGG [30] and ResNet [31], as well as the recent lightweight ShuffleNet series [32], [33], MobileNet series [34], [35] and GhostNet [20]. Table I presents their performance on the ImageNet dataset along with the number of parameters and floating point operations (i.e., FLOPs). Considering the trade-off between efficiency and accuracy, this paper uses GhostNet as the CNN for image feature extraction.

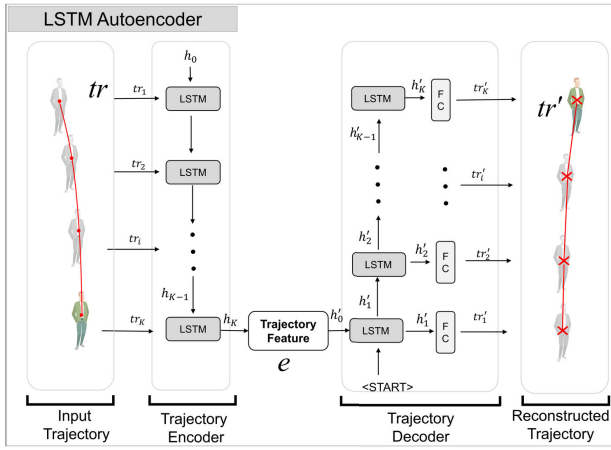


Fig. 4. Model architecture of the LSTM autoencoder, which consists of a trajectory encoder and a decoder for trajectory feature extraction in an unsupervised learning manner.

2) *Trajectory Feature Extractor*: Trajectories are a type of structured data of varying length, which need to be processed into trajectory features of equal length before inputting to the intention prediction decoder. As outlined in [10], Long Short-Term Memory (LSTM) autoencoder is capable of extracting features (such as direction and speed) underlying trajectories automatically. To this end, we devise an LSTM autoencoder consisting of a trajectory encoder and a decoder for unsupervised trajectory feature extraction, as illustrated in Figure 4.

In the LSTM autoencoder, the input of the encoder is a sorted set of trajectory points $tr = \{tr_1, \dots, tr_K\}$ in the form of pixel coordinates, where $tr_t = (x_t, y_t)$. (x_t, y_t) is the pixel coordinates of the observed pedestrian at time t . The decoder takes the encoder's hidden state h_K as input and progressively produces a reconstructed trajectory $tr' = \{tr'_1, \dots, tr'_K\}$ after being processed by a fully-connected layer. Formally, suppose that the operations of trajectory encoder and decoder are denoted as $f_E(\cdot; w_e)$ and $f_D(\cdot; w_d)$, respectively. The training objective of the LSTM autoencoder could be formulated as:

$$L_{TFA} = \|tr' - tr\|_F^2 = \frac{1}{n} \sum_{t=1}^K \|tr'_t - tr_t\|^2$$

$$tr' = f_D(e; w_d)$$

$$e = f_E(tr; w_e) \quad (10)$$

The loss function L_{TFA} serves to ensure that the reconstructed trajectory is consistent with the input trajectory, thereby enabling the learned trajectory feature e to accurately represent the original trajectory of varying length. In our experiments, we utilize all pedestrian trajectories extracted by an object tracker (i.e., YOLO V5 [37] in conjunction with DeepSORT [38]) to train the LSTM autoencoder. We choose DeepSORT as it can effectively alleviate the pedestrian re-identification problem in obscured/occluded cases, improving the continuity of pedestrian trajectories. After the autoencoder reaches convergence, the encoder is used as the trajectory feature extractor while the decoder is detached.

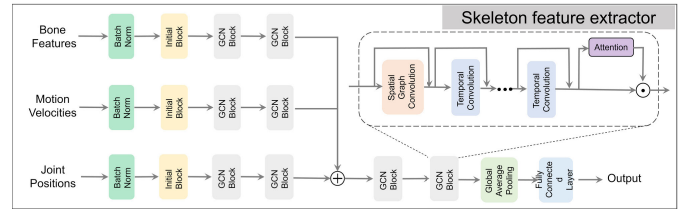


Fig. 5. Model Architecture of the skeleton feature extractor, which models the action information underlying pedestrians' skeletons into the framework and is expected to improve the performance on pedestrian crossing intention prediction.

3) *Skeleton Feature Extractor*: The dynamics of pedestrian body skeletons convey a wealth of information about pedestrian actions, which is also crucial for predicting their crossing intentions. In other words, the skeleton sequences of pedestrians tending to cross the road may embrace actions such as leg lifting or waving, which are generally different from those of pedestrians having no intention to cross. As such, modeling the action information underlying pedestrians' skeletons into the framework is expected to improve the performance of pedestrian crossing intention prediction.

In this study, we first adopt an advanced human pose estimator dubbed as HRNet [39] to capture the pedestrians' body skeletons $\mathcal{X} = \{x \in \mathbb{R}^{C \times T \times V}\}$ in surveillance scenarios, where C , T , and V denote the coordinate, frame and joint, respectively. In cases where pedestrians are temporarily occluded, their body skeletons during the occlusion period are estimated by interpolating the skeletons before and after the occlusion. If pedestrians remain occluded for a prolonged duration, the coordinates of their unrecognized joint points are set to 0, so as to enable the subsequent network to disregard them. The body skeletons undergo preprocessing to generate three classes of node features, i.e., joint positions $\mathcal{P} = \{p_i | i = 1, \dots, V\}$, motion velocities $\mathcal{M} = \{m_i | i = 1, \dots, V\}$ and bone features $\mathcal{B} = \{(l_i, a_i) | i = 1, \dots, V\}$, where,

$$p_i = x[:, :, i] - x[:, :, c]$$

$$m_i = x[:, t+1, :] - x[:, t, :], t = 1, 2, \dots, T$$

$$l_i = x[:, :, i] - x[:, :, i_{adj}]$$

$$a_{i,w} = \arccos\left(\frac{l_{i,w}}{\sqrt{l_{i,x}^2 + l_{i,y}^2}}\right) \quad (11)$$

In the formula above, p_i represents the normalized position features and c denotes the index of the center spine joint; m_i represents the motion velocities of all joints between two adjacent frames; l_i denotes the bone lengths and $a_{i,w}$ denotes the bone angles, where i_{adj} means the adjacent joint of the i -th joint and $w \in (x, y)$ is the 2D coordinates.

After preprocessing the skeleton data, a GCN-based skeleton feature extractor following [40] is designed, whose detailed architecture is demonstrated in Figure 5. Specifically, three types of node features are respectively fed to three different network branches, in each branch there is a batch normalization layer for fast convergence, an initial block for data-to-feature transformation and two GCN blocks for

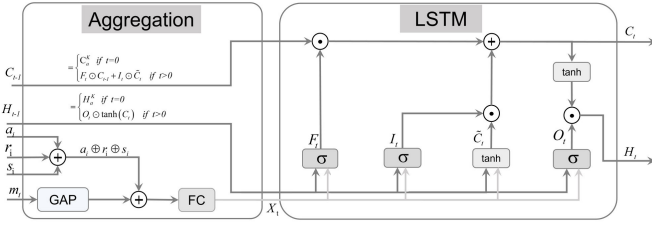


Fig. 6. Model architecture of the intention prediction decoder, which aggregates the features extracted by the environment encoder and the pedestrian-state encoder, and outputs the crossing intention possibility of the observed pedestrian.

informative feature extraction. After that, the processed inputs are fused by a concatenation operation. Eventually, the fused feature is further processed by two additional GCN blocks, followed by a global averaging pooling layer for feature abstraction and a fully connected layer for classification output. Notably, in each GCN block, there is one spatial graph convolution and several temporal convolutions followed by an attention module, where the attention module is designed for distinguishing the most informative joints and more valuable frames.

In this study, the skeleton feature extractor is pre-trained on the Kinetics 400 dataset [41], which covers a broad range of classes related to human actions. Thereafter, the last fully connected layer is discarded and the rest is used for feature extraction of pedestrian skeleton sequences.

F. Intention Prediction Decoder

Last, an LSTM-based intention prediction decoder is proposed to aggregate the features extracted by the environment encoder and the pedestrian-state encoder, and outputs the crossing intention possibility of the observed pedestrian. Suppose that (i) the outputs of environment encoder are denoted as (C_a^K, H_a^K) of the GCN-based autoencoder and $M = \{m_1, m_2, \dots, m_K\}$ of ST-GCNN, and (ii) the outputs of pedestrian-state encoder are denoted as $\{a_1, a_2, \dots, a_K\}$ of the appearance feature extractor, $\{r_1, r_2, \dots, r_K\}$ of the trajectory feature extractor and $\{s_1, s_2, \dots, s_K\}$ of the skeleton feature extractor. The aggregation and reasoning process of the intention prediction decoder could be found in Figure 6 and mathematically formulated as:

$$X_t = f_{FC}(GAP(m_t) \oplus (a_t \oplus r_t \oplus s_t); W_{FC}) \quad (12)$$

$$H_t = \begin{cases} H_a^K & \text{if } t = 0 \\ O_t \odot \tanh(C_t) & \text{if } t > 0 \end{cases} \quad (13)$$

$$C_t = \begin{cases} C_a^K & \text{if } t = 0 \\ F_t \odot C_{t-1} + I_t \odot \tilde{C}_t & \text{if } t > 0 \end{cases} \quad (14)$$

$$\tilde{C}_t = \tanh(X_t W_{xc} + H_{t-1} W_{hc} + b_c) \quad (15)$$

$$I_t = \sigma(X_t W_{xi} + H_{t-1} W_{hi} + b_i) \quad (16)$$

$$F_t = \sigma(X_t W_{xf} + H_{t-1} W_{hf} + b_f) \quad (17)$$

$$O_t = \sigma(X_t W_{xo} + H_{t-1} W_{ho} + b_o), \quad (18)$$

where I_t , F_t , O_t , C_t , and H_t are input gate, forget gate, output gate, memory cell, and output response in the LSTM network; functions f_{FC} and GAP represent the operations of fully

connected layer and global average pooling layer, and $\tanh$ and σ denote tanh and sigmoid activation functions. Notably, the learnable parameters are denoted as W (e.g., W_{xc}) and b (e.g., b_c), the symbols $\oplus$ and $\odot$ represent element-wise addition and multiplication.

IV. EXPERIMENTAL RESULTS

A. Dataset

To our best understanding, there are limited customized datasets available for pedestrian crossing intention prediction from surveillance videos. This is primarily due to several complex factors that need to be considered, such as privacy concerns and scenario intricacies. As previously mentioned, predicting pedestrian crossing intention from surveillance cameras is capable of timely transmitting the over-the-horizon early warnings to approaching vehicles, compared to that from on-board cameras. This motivates us to develop a novel dataset for surveillance camera-based crossing intention prediction.

Specifically, we first collect a large amount of video data (over 127 hours) from surveillance cameras located in the urban roads of Jiangsu and Zhejiang provinces (China). Next, pedestrians are detected and tracked by YOLO V5 [37] plus DeepSORT [38] trained on SEU_PML dataset [42]. To obtain more accurate pedestrian trajectories, the trajectories extracted by YOLO V5 plus DeepSORT are further manually rectified since there may be non-negligible ID switches. After that, all pedestrians are divided into two categories (i.e., crossing and no crossing) after carefully observing their behaviors.

To attain positive samples for model training, we crop the videos involving pedestrian crossing into video clips of duration K_1 ($K_1 = 1s$), ending at the time of K_2 seconds before the observed pedestrian starts to cross, as presented in Figure 7. More concretely, if a pedestrian i begins to cross at time t_i , the truncated video clip would span from $t_i - K_1 - K_2$ seconds to $t_i - K_2$ seconds. Notably, K_2 varies from 0 to 3 seconds to enable an investigation of how early the model could predict the pedestrian's intention. In this context, "crossing intention prediction" in this study refers to judging whether the observed pedestrian attempts to cross the road K_2 seconds later, after watching a video clip of duration K_1 . Regarding the collection of negative samples, we randomly crop videos containing pedestrians not crossing to video clips of duration K_1 .

After we process all the collected videos using a publicized annotation tool named CVAT,¹ a total of 7532 pedestrians, including 2892 and 4640 pedestrians with and without crossing behavior, are captured and each of them is assigned a unique tracking ID. Note that we discard those crossing pedestrians with insufficient waiting time (i.e., less than $K_1 + K_2$ seconds) and non-crossing pedestrians staying for a short time (i.e., less than K_1 seconds) since they cannot constitute a complete sample for model training. After careful selection, A total of 6436 positive samples (i.e., 1935+1726+1554+1221) and 2809 negative samples are obtained by traversing K_2 from 0 to 3 seconds at 1-second intervals. In our experiments, we divide the whole samples into a training set (4505 positives/2266

¹<https://github.com/openvinotoolkit/cvat>

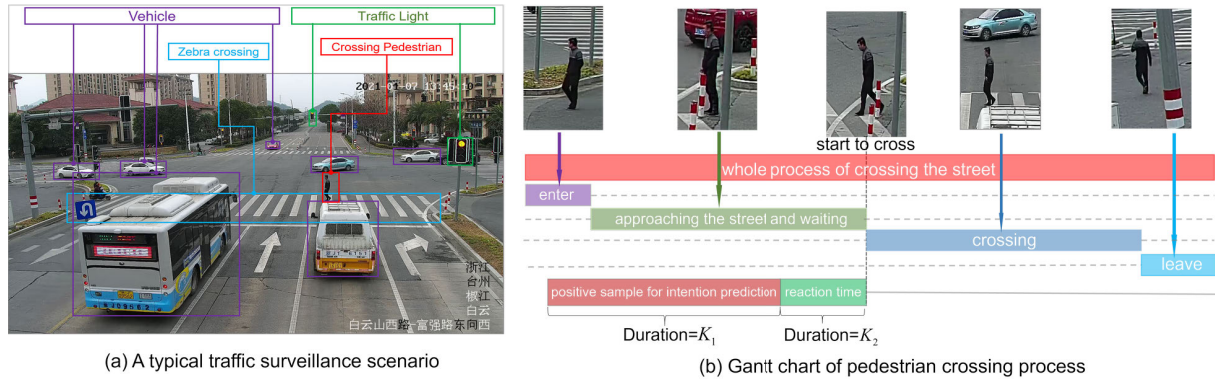


Fig. 7. (a) A typical traffic surveillance scenario containing crossing pedestrians; (b) gantt chart depicting the pedestrian crossing process. Notably, in order to attain positive samples for model training, we crop the videos involving pedestrian crossing into video clips of duration K_1 ($K_1 = 1s$), ending at the time of K_2 seconds before the observed pedestrian starts to cross.

negatives), a validation set (644 positives /181 negatives), and a test set (1287 positives/362 negatives) with the ratio of 70%, 10% and 20%, respectively.

B. Experimental Setting

In this study, all experiments are implemented on a workstation with a NVIDIA RTX 3090 GPU and an Intel (R) Core i9-10900X CPU, and all code is written by Pytorch 1.8.0 and Python 3.8.2.

Regarding the appearance sub-graph, we take into account the appearance variation of five vehicles, three pedestrians and two traffic signals nearby the observed pedestrian. In this case, M , N , and L are assigned values of 5, 3, and 2, respectively. To note, if the number of instances (i.e., vehicle, pedestrian, and traffic light) nearby the observed pedestrian is less than the default value, we will pad zeros to enable the framework to learn to ignore those missing instances. This allows the proposed framework to work effectively in traffic scenes having no traffic light (e.g., unsignalized road sections).

Regarding the motion sub-graph, apart from five vehicles and three pedestrians nearby, two nearest zebra crossings are taken in consideration as well, since a pedestrian standing at an intersection could cross the street in either a longitudinal or transverse direction. Consequently, the parameter Z in the motion sub-graph is set to 2.

To extract representative features from the appearance sub-graph, we use a four-layer GCN autoencoder that maps each node representation from $(1 \times 1280) \rightarrow (1 \times 512) \rightarrow (1 \times 256) \rightarrow (1 \times 512) \rightarrow (1 \times 1280)$. Notably, each graph convolutional layer is accompanied by a ReLU activation, trained with a learning rate of 0.001 and a weighting coefficient γ of 0.5. Moreover, a single layer of STGCN, which operates one graph convolution and three temporal convolutions, is adopted to extract motion features reflecting spatiotemporal relationships between pedestrians and their surroundings.

Regarding the implementation of the pedestrian-state encoder, we adopt a lightweight model named GhostNet² for image feature extraction, in which the last fully connected layer for classification output is discarded. To extract semantic

representation from trajectories, an LSTM autoencoder is pre-trained in an unsupervised manner, by taking multiple pedestrians' trajectories extracted from surveillance videos as both input and annotation. Some hyper-parameters for LSTM autoencoder training are set as follows: training epoch and batch size are set to 100 and 256, and an Adam optimizer with initial learning rate of 0.001 is used to update the autoencoder's parameters. When the LSTM autoencoder converges, the LSTM encoder therein will be retained to extract trajectory features, while the decoder will be discarded. Following the recent publicized work,³ we build our skeleton feature extractor pre-trained on the Kinetics 400 dataset using an SGD optimizer with initial learning rate of 0.1 and momentum of 0.9.

Regarding the implementation of the intention prediction decoder, a single layer of LSTM network is adopted to aggregate the features extracted by both the environment encoder and the pedestrian-state encoder. The input size and the hidden unit size of this LSTM is set to 256 and 64, respectively. It should be noted that the proposed framework achieves performance convergence over a total of 5 days and 16 hours of training on our experimental workstation.

As outlined in Section III-A, pedestrian crossing intention prediction refers to anticipating whether a pedestrian i has an intention of crossing the street $int_i \in \{0, 1\}$ after a short period of observation of both pedestrians themselves and their surroundings. Therefore, we define the task as a binary classification problem. In this study, we adopt the widely used binary cross entropy loss as the overall learning objective to update the learnable parameters of our framework.

Regarding the evaluation metric, we adopt precision, recall and accuracy to measure the performance, which are defined as follows:

$$\begin{aligned}
 precision &= \frac{TP}{TP + FP} \\
 recall &= \frac{TP}{TP + FN} \\
 accuracy &= \frac{TP + TN}{TP + TN + FP + FN}, \quad (19)
 \end{aligned}$$

²<https://github.com/iamhankai/ghostnet.pytorch>

³<https://github.com/yfsong0709/EfficientGCNv1>

where TP , FP and FN represent true positive, false positive and false negative, respectively. Additionally, we introduce the precision-recall curve along with AUC value (short for Area Under the Curve) to compare our method with other state-of-the-arts.

C. Result and Discussion

1) *Ablation Study*: To examine the importance of each component proposed in the framework, three groups of ablation experiments are conducted separately from the perspectives of pedestrian-only, environment-only, and a combination of both (i.e., pedestrian+ environment). For clarity, the methods adopted in the three groups of ablation experiments are denoted as pedestrian-only, environment-only, and pedestrian-environment methods, respectively.

In the pedestrian-only ablation experiments, we introduce several typical methods for action recognition (e.g., Conv-LSTM [43] and 3DCNN [44]) to compare the prediction performance, in addition to ablating components (i.e., GhostNet [20], LSTM [45], and EfficientGCN [40]) in the pedestrian state encoder. Concretely, the Conv-LSTM method first adopts a CNN model for feature extraction followed by a series of LSTM for sequence learning. The 3DCNN method extends 2D convolutions into 3D ones, in which an extra dimension is used to aggregate temporal features. In the environment-only ablation experiments, we report the results of ablating the GCN-based autoencoder and ST-GCNN from the environment encoder (i.e., GCN-based autoencoder+ST-GCNN), respectively. Finally, we report the performance of the overall framework, which takes as inputs not only three types of pedestrian states but also two pedestrian-centric environment sub-graphs. To note, we report all results using the accuracy metric and we vary the reaction time left for approaching vehicles (i.e., K_2) from 0 seconds to 3 seconds to assess how early those methods are able to anticipate a crosser.

2) *Discussion*: As demonstrated in Table 2, it can be easily observed that the combination of GhostNet, LSTM, and EfficientGCN (i.e., the pedestrian state encoder) attains the best performance among all pedestrian-only methods. Furthermore, ablating any component will degrade the prediction performance. For example, when $K_2 = 3s$, ablating the appearance feature extractor (i.e., GhostNet) will reduce the accuracy by 1.8%, while ablating the skeleton feature extractor (i.e., EfficientGCN) will reduce by 4.4%. This implies that each component in our proposed pedestrian state encoder can complement and reinforce each other. Another point to note is that the EfficientGCN method processing skeleton sequence performs better than those methods processing other individual input type, such as trajectory or image sequence. A reasonable explanation is that skeleton sequence could highly abstract pedestrian behavior, which lends it an advantage over image sequence that contains richer yet less refined information.

In all environment-only methods, it should be noted that the combination of GCN-based autoencoder and ST-GCNN outperforms either of them, achieving an accuracy of 72.4% when $K_2 = 3s$. This indicates that both the appearance variations and the spatiotemporal relationships between pedestrians



Fig. 8. Some representative detection results. Each detection result therein is demonstrated by an image overviewing the entire traffic scene and a series of image crops focusing on the observed pedestrian. It is worth noting that, in each image crop, we report the crossing intention intensities predicted by three different methods, i.e., pedestrian-only (GhostNet+ LSTM+ EfficientGCN, abbreviated as P), environment-only (GCN-based autoencoder+ST-GCNN, abbreviated as E) and pedestrian-environment (Ours, abbreviated as P+E) methods.

and their surroundings play an important role for pedestrian crossing intention prediction, and their combination endows the framework with a stronger ability to model various environmental factors.

As a combination of all the superior methods above, our method achieves the highest detection accuracy when K_2 varies from 0 seconds to 3 seconds, indicating its effectiveness and superiority. An interesting finding is that the prediction performance drops slightly when K_2 gets larger. This is in line with common sense, because the closer pedestrians are to the moment they start to cross, i.e., the smaller K_2 , and the more apparent their crossing intention will be. To sum up, our method is able to early predict the crossing intention for those pedestrians who are about to cross the street three seconds later, with an acceptable accuracy of 75.0%, which is of practical significance for preventing traffic injuries and improving pedestrian safety.

3) *Visualization*: For a more intuitive understanding of how the proposed framework outperforms pedestrian-only or environment-only methods, we present some representative detection results in Figure 8. For clarity, each detection result is demonstrated by an image overviewing the entire traffic scene and a series of image crops focusing on the observed pedestrian. It is worth noting that, in each image crop, we report the crossing intention intensities predicted by three different methods, i.e., pedestrian-only (GhostNet+ LSTM+ EfficientGCN, abbreviated as P), environment-only

TABLE II

ABLATION STUDY OF THE PROPOSED FRAMEWORK, IN WHICH THE RESULTS ARE EVALUATED AND COMPARED USING THE ACCURACY METRIC

Category	Model	Input type	$K_2 = 0s$	$K_2 = 1s$	$K_2 = 2s$	$K_2 = 3s$
pedestrian-only	GhostNet	image	59.2	57.7	56.8	55.5
	LSTM	trajectory	56.6	55.2	55.3	53.9
	Conv-LSTM	image sequence	63.3	62.5	62.3	60.9
	3DCNN	image sequence	62.0	60.4	60.2	59.7
	EfficientGCN	skeleton sequence	64.3	64.0	63.5	62.6
	GhostNet + LSTM	image + trajectory	64.8	63.2	62.6	61.3
	GhostNet + EfficientGCN	image + skeleton	65.2	64.5	64.2	63.1
	LSTM + EfficientGCN	trajectory + skeleton	66.8	66.2	65.7	63.9
	GhostNet + LSTM + EfficientGCN	image + trajectory + skeleton	67.6	67.1	66.5	65.7
environment-only	GCN-based autoencoder	appearance sub-graph	66.2	66.5	65.4	63.8
	ST-GCNN	motion sub-graph	71.8	70.7	70.6	69.9
	GCN-based autoencoder + ST-GCNN	appearance sub-graph + motion sub-graph	73.6	73.2	73.3	72.4
pedestrian-environment	Ours	image + trajectory + skeleton + two subgraphs	76.8	76.1	75.7	75.0

(GCN-based autoencoder+ ST-GCNN, abbreviated as E) and pedestrian-environment (Ours, abbreviated as P+E) methods.

In Scene 1 and Scene 2 of Figure 8, the observed pedestrians both walked towards the zebra crossing, but in Scene 1, the pedestrian directly crossed the road without thinking, while in Scene 2 the pedestrian waited for a while before crossing the road. In Scene 3, the observed pedestrian passed by the zebra crossing but did not cross the road. In Scene 4, we provide a hard negative, in which a man walked to the zebra crossing, thought for a moment and then left without crossing the road. In Scene 5, a woman crossed the road while watching the traffic on an unsignalized road section. Scene 6 presents that a man took one look at the road and began to cross in a nighttime condition.

It can be seen that the proposed framework (i.e., P+E) exhibits the highest intention intensities for the observed pedestrians in both Scene 1 and Scene 2 as well as the lowest intention intensity in Scene 3, indicating its superior discrimination ability. However, our framework fails in Scene 4 as the observed pedestrian exhibits similar behavior to those attempting to cross, such as waiting for a while nearby the zebra crossing. This implies that our framework still has room to improve, especially for those hard negatives. The results in Scene 5 and 6 suggest that the proposed framework is also suited for pedestrian crossing intention prediction in both unsignalized traffic scenes and nighttime conditions.

D. Comparison With State-of-the-Arts

To demonstrate the superiority of our framework, this study compares performance with existing state-of-the-art models designed for pedestrian crossing intention prediction based on on-board cameras. Comparison results are demonstrated in Figure 9 (a-d), in the form of precision-recall curves and AUC values. It should be noted that we modulate those methods to make them more suitable for surveillance scenarios on the premise that their core ideas remain unchanged. Specifically, those models along with their descriptions are as follows:

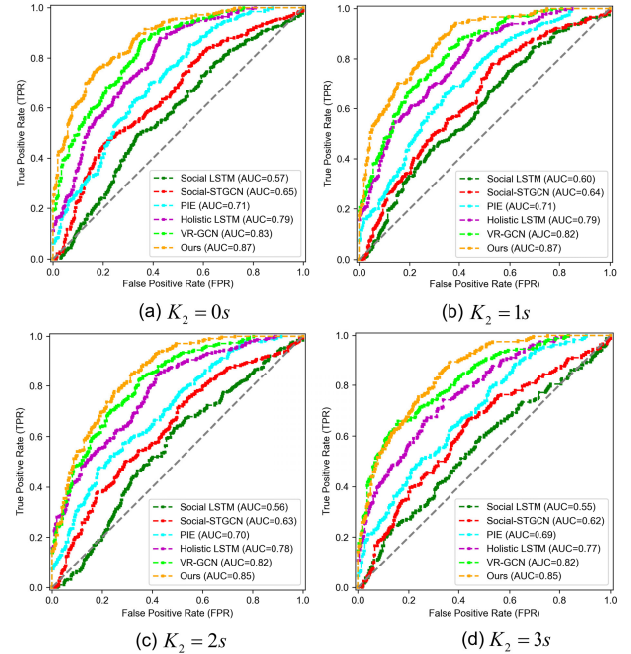


Fig. 9. Comparison results with other state-of-the-arts, in the form of precision-recall curves along with AUC values.

(1) **Social LSTM** [4]: which predicts pedestrians' future trajectories using a series of LSTM networks and social pooling layers to model the social relationships among pedestrians. In this study, the crossing intention is judged according to whether the predicted trajectory intersects with the area of zebra crossing or road.

(2) **Social-STGCNN** [13]: which predicts pedestrians' future trajectories models based on a spatiotemporal graph convolutional neural network that models the pedestrians' interactions as a graph. Also, utilizing it for predicting crossing intentions is akin to the approach employed by **Social LSTM**.

(3) **PIE** [2]: which receives as input a sequence of images, bounding boxes and the current speed of the ego-vehicle.

Thereinto, the image sequence and bounding box sequence are processed by a series of Conv-LSTMs and LSTMs, respectively. The ego-vehicle speed is processed by an encoder-decoder scheme followed by a series of self-attention units. Due to the unavailability of the ego-vehicle speed in our study, we utilize the relative speeds between the observed pedestrian and its nearest five vehicles instead.

(4) **Holistic LSTM** [8]: which incorporates multiple information cues from pedestrians and vehicles, improving the transferability of LSTMs in modeling future variations through introducing a speed cell, an intention cell and a correlation cell. Notably, we remain optical flow yet discard depth estimation in the conventional scheme of **Holistic LSTM**, since it is non-trivial to achieve depth estimation from surveillance cameras.

(5) **VR-GCN** [3]: which develops graph convolutional autoencoders to abstract visual features between pedestrians and their surrounding objects, and then incorporates pedestrian bounding boxes and human poses into the LSTM-based prediction module to reason pedestrians' crossing intention.

As presented in Figure 9, it can be seen that our proposed method surpasses all existing models for pedestrian intention detection, attaining state-of-the-art performance. Among all models, **Social LSTM** and **Social-STGCNN** perform worst since only the trajectory information of instances in surveillance scenarios are received as input in the two models. Conversely, **PIE**, **Holistic LSTM** and **VR-GCN** incorporate the appearance information of pedestrians, thus achieving higher AUC values than **Social LSTM** and **Social-STGCNN**. Nevertheless, those methods are designed for pedestrian crossing intention prediction from on-board cameras, which do not deeply explore the complicated pedestrian-environment relationships in surveillance scenarios, and thus reach saturated accuracy. In conclusion, the performance improvement over those existing state-of-the-arts indicates the superiority of our proposed method for spatiotemporal modeling and reasoning as well as crossing intention prediction from surveillance cameras, which is expected to be applied for over-the-horizon safety warning in future intelligent transportation systems.

E. Inference Speed Analysis

In this section, we discuss the inference speed of the proposed framework. First, we assume that the inputs (i.e., trajectories, appearance sequences, and skeleton sequences) of the proposed framework are off-the-shelf, e.g., acquired in real time from roadside computing devices. In this case, the average processing time per frame is approximately 0.031 seconds. Consequently, our framework can achieve an inference speed of 32 frames per second, thereby satisfying the real-time requirements. If the time of input acquisition (e.g., detection, tracking and pose estimation) is taken into consideration, it should be admitted that our framework may not meet the real-time requirements. In the future, with the advancements of hardware (e.g., higher computing power) and algorithms (e.g., model quantization [46] and pruning [47]), we believe that the inference speed of our framework is expected to reach a real-time level, even if detection, tracking, and pose estimation are also considered.

Another important point to note is that the proposed framework could forecast pedestrian crossing intentions up to three seconds in advance, which could greatly loosen the real-time requirements of our framework. In other words, we believe that the total time of input acquisition, model inference, information delivery and driver reaction is less than 3s, the framework could still be considered effective. As far as we know, the communication delays from the cloud to the connected vehicle in the 5G environment are usually less than 10ms [48], and the reaction times of a general driver range between 0.4 to 0.6s [49]. This means that our framework can tolerate more than 2s spent on input acquisition, which is sufficient for existing detection, tracking and pose estimation models.

V. CONCLUSION

This paper focuses on pedestrian crossing intention prediction from surveillance cameras, which could provide over-the-horizon safety warnings and has the potential to better ensure pedestrian safety. To handle such a challenging task, we propose an automated learning framework, in which a pedestrian-centric environment graph is primarily constructed to reflect visual variations and spatiotemporal relationships between pedestrians and their surroundings. Then, a GCN-based environment encoder and a pedestrian-state encoder are designed to extract prominent environment features and pedestrian behavior features, respectively. Last, an intention prediction decoder is developed to extrapolate the probability of crossing intention. Experimental results demonstrate that our proposed framework attains state-of-the-art performance, suggesting its effectiveness and superiority. Based on the predicted pedestrian crossing intention, traffic management departments could develop corresponding active safety warning systems, which adopt some warning measures (e.g., flashing in-curb LEDs warning and on-board unit warning) to timely send pedestrian crossing intentions to relevant vehicles and alert the inattentive drivers. This would be conducive to further enhancing pedestrian crossing safety.

Although our framework has attained a desired result, it still has great room for improvement, especially in handling those hard negatives (e.g., pedestrians close to the curb but having no intention to cross). Besides, this paper does not provide the prediction results in adverse weather conditions (e.g., snowy and foggy) due to the scarcity of related videos and the difficulty of annotation. In the future, we will attempt to collect more samples in adverse weather conditions and continuously update our framework to a more generic one.

In addition, pedestrian intentions are divided into two categories in this paper, namely, crossing and non-crossing. In the future, we will also consider dividing pedestrian intentions into more fine-grained categories, such as separately distinguishing pedestrians who change their decision to cross due to the impact of environmental factors. This will be more conducive to optimizing the active safety intervention measures at cross-walks of intersections and road sections.

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